



NSW Education Standards Authority

2019 HIGHER SCHOOL CERTIFICATE EXAMINATION

Food Technology

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen

**Total marks:
100****Section I – 20 marks** (pages 2–6)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 50 marks (pages 9–17)

- Attempt Questions 21–26
- Allow about 1 hour and 25 minutes for this section

Section III – 15 marks (page 33)

- Attempt Question 27
- Allow about 30 minutes for this section

Section IV – 15 marks (page 34)

- Attempt Question 28
- Allow about 30 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** A food manufacturing company operating in many countries would be best classified as a

 - A. large company.
 - B. small business.
 - C. multinational.
 - D. household.

- 2** Which sector of the Australian food industry is significantly affected by severe drought?

 - A. Food retail
 - B. Agriculture and fisheries
 - C. Food service and catering
 - D. Food processing/manufacturing

- 3** What are the ‘four Ps’ of a food marketing plan?

 - A. Product, price, place and publicity
 - B. Product, price, place and promotion
 - C. Produce, price, place and promotion
 - D. Produce, price, position and promotion

- 4** Which of the following is an example of a food where its production involves pasteurisation?

 - A. Cheese
 - B. Fresh seafood
 - C. Savoury biscuits
 - D. Frozen cauliflower

- 5 An example of a community group that promotes health in Australian society is
- A. CSIRO.
 - B. FSANZ.
 - C. Meals on Wheels.
 - D. NSW Food Authority.
- 6 Which of the following is the most significant benefit of using organic farming systems?
- A. Less landfill
 - B. Cleaner waterways
 - C. Reduced cost to the consumer
 - D. Reduced cost to the manufacturer
- 7 Food sold in retail stores in Australia must comply with Country of Origin labelling laws that were introduced in 2016.
- These labelling laws are most likely to benefit
- A. local councils.
 - B. import markets.
 - C. primary producers.
 - D. people with allergies.
- 8 Which is an example of unethical food advertising?
- A. Bundle pricing
 - B. A celebrity endorsement
 - C. Exaggerated nutritional claim
 - D. Marketing through cross-promotion

- 9** A food company is considering introducing a line extension to their fresh pasta range.

Where would this be represented in a SWOT analysis?

- A. Threat
- B. Strength
- C. Weakness
- D. Opportunity

- 10** Which of the following is a result of enzymatic activity?

- A. Fruit becoming over-ripe
- B. Mould developing in jam
- C. Flour infested by weevils
- D. Savoury biscuits becoming stale

- 11** Virtual supermarkets enable consumers to purchase grocery items by scanning barcodes with their smartphone.

What is this an example of?

- A. Product placement
- B. Emerging technology
- C. A food marketing plan
- D. A value-added experience

- 12** At which step of the food product development process would a beverage company conduct a sensory evaluation?

- A. Market research
- B. Feasibility study
- C. Testing product prototype
- D. Production process development

- 13** Which of the following includes ONLY functional foods?
- A. High-calcium milk, almonds, table margarine
 - B. High-fibre breakfast drinks, brown rice, probiotic yoghurt
 - C. Breakfast cereals containing folate, omega-3 bread with soy and linseed, organic apples
 - D. Breakfast cereals containing prebiotic barley, orange juice containing iron, hi-maize muffins
- 14** In many religions people abstain from certain foods and have specific periods of fasting. What is this an example of?
- A. Lifestyle and dietary practices
 - B. Social and nutritional practices
 - C. Nutritional and dietary practices
 - D. Traditional and cultural practices
- 15** Product specifications for the development of a new food product are an essential component of the design brief because they
- A. provide the guidelines within which the developer must work.
 - B. enable the developer to have flexibility in the design outcome.
 - C. include information about the considerations of the project but not any outcomes.
 - D. include information about the constraints of the project but not any considerations.
- 16** A food company is looking to extend their beverage line by producing a fermented probiotic drink. The target market for this product is most likely to be
- A. all consumers.
 - B. a specific group of consumers.
 - C. consumers in a particular age group.
 - D. a group of consumers of the same gender.

- 17** Which combination of medical conditions is most significantly related to cardiovascular disease?
- A. Atherosclerosis and stroke
 - B. Type 2 diabetes and anaemia
 - C. Diverticulitis and osteoporosis
 - D. Hypertension and dental caries
- 18** A multi-pack of canned vegetables has been boxed and shrink-wrapped for distribution.
- This is an example of
- A. tetra packaging.
 - B. tertiary packaging.
 - C. primary packaging.
 - D. modified atmosphere packaging.
- 19** Which food additive is used by manufacturers to reduce moisture loss in foods?
- A. Acids
 - B. Dextrins
 - C. Emulsifiers
 - D. Humectants
- 20** Which piece of equipment would a commercial manufacturer use to remove pulp from orange juice?
- A. Sieve
 - B. Retort
 - C. Centrifuge
 - D. Blast chiller

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Centre Number

Food Technology

Sections II, III and IV

Answer Booklet

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Student Number

Instructions

- Write your Centre Number and Student Number at the top of this page
- Use this answer booklet for Sections II, III and IV
- Follow the instructions inside this booklet as to where you should write your answers

Please turn over

Section II

50 marks

Attempt Questions 21–26

Allow about 1 hour and 25 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (5 marks)

- (a) Identify TWO reasons for developing a new food product.

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- (b) Complete the following table.

3

<i>Type of food product development</i>	<i>Food product example</i>
Me too	
Line extension	
New to the world	

Do NOT write in this area.

Question 22 (11 marks)

- (a) Other than environmental concerns, outline TWO issues for consumers related to food packaging. **2**

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- (b) Describe how the food retail sector has responded to consumer concerns about the environment. **3**

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Question 22 continues on page 12

Question 22 (continued)

- (c) Explain current legislative requirements related to labelling food in Australia. **6**

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End of Question 22

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Question 23 (8 marks)

- (a) Outline nutritional considerations for a specific group within Australian society. **3**

Group:

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- (b) Explain a specific strategy that could improve the health outcomes of a group you have investigated. In your answer, you may refer to the group you named in part (a) or a different group. **5**

Group:

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Question 24 (9 marks)

- (a) Identify gender issues in ONE sector of the Australian food industry. 2

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- (b) Describe how an organisation within the Australian food industry has responded to lifestyle changes within society. 3

Organisation:

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Question 24 continues on page 15

Do NOT write in this area.

Question 24 (continued)

- (c) How has an organisation you have studied affected the Australian economy? **4**
Include relevant examples in your answer.

Organisation:

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End of Question 24

Question 25 (9 marks)

- (a) An Australian food manufacturer has recently developed a gluten-free sourdough bread.

(i) Identify a potential target market for this product.

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(ii) Justify your choice of target market.

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- (b) Examine the use of genetic modification in the Australian food industry. Include relevant examples in your answer.

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Question 26 (8 marks)

- (a) Define the term *malnutrition*.

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- (b) Discuss the role of dietary supplements in an individual's diet.

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Section III — Answer Question 27 on pages 18–24

27

Question Number

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If you require more space to answer Section III, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section III, tick here.

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Section IV — Answer Question 28 on pages 25–31

28

Question Number

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If you require more space to answer Section IV, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section IV, tick here.

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Food Technology

Section III

15 marks

Attempt Question 27

Allow about 30 minutes for this section

Answer the question on pages 18–24 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Question 27 (15 marks)

In 2018, a small number of strawberries in Australia were physically contaminated, affecting supply and consumption. Due to safety risks to consumers, strawberries were removed from sale.

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|--|----------|
| (a) Outline how TWO sectors of the Australian food industry were affected by this event. | 3 |
| (b) Describe quality management considerations that may have been applied in this food safety incident. | 4 |
| (c) Explain TWO preservation processes that could have been adopted to assist in minimising the waste from the excess supply of strawberries. Include relevant examples in you answer. | 8 |

Please turn over

Section IV

15 marks

Attempt Question 28

Allow about 30 minutes for this section

Answer the question on pages 25–31 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
 - apply course concepts to food technology issues
 - communicate ideas and information using appropriate terminology and relevant examples
 - present a logical and cohesive response
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Question 28 (15 marks)

Explain how food manufacturers have responded to consumer concerns about health and diet related disorders.

End of paper